

Jetson CUDA Practice Env

실습을 위한 기초 환경

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00 What We are Doing

- In RDA(실시간 데이터 해석) course,
- We are preparing CUDA coding practice.
- We need real time data (streaming, in sequence)

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NVIDIA Jetson Nano

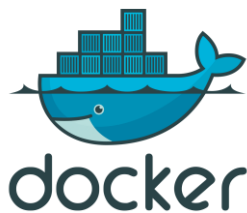
```
nvcc --version #command to confirm nvidia gpu
```

00 What We are Doing

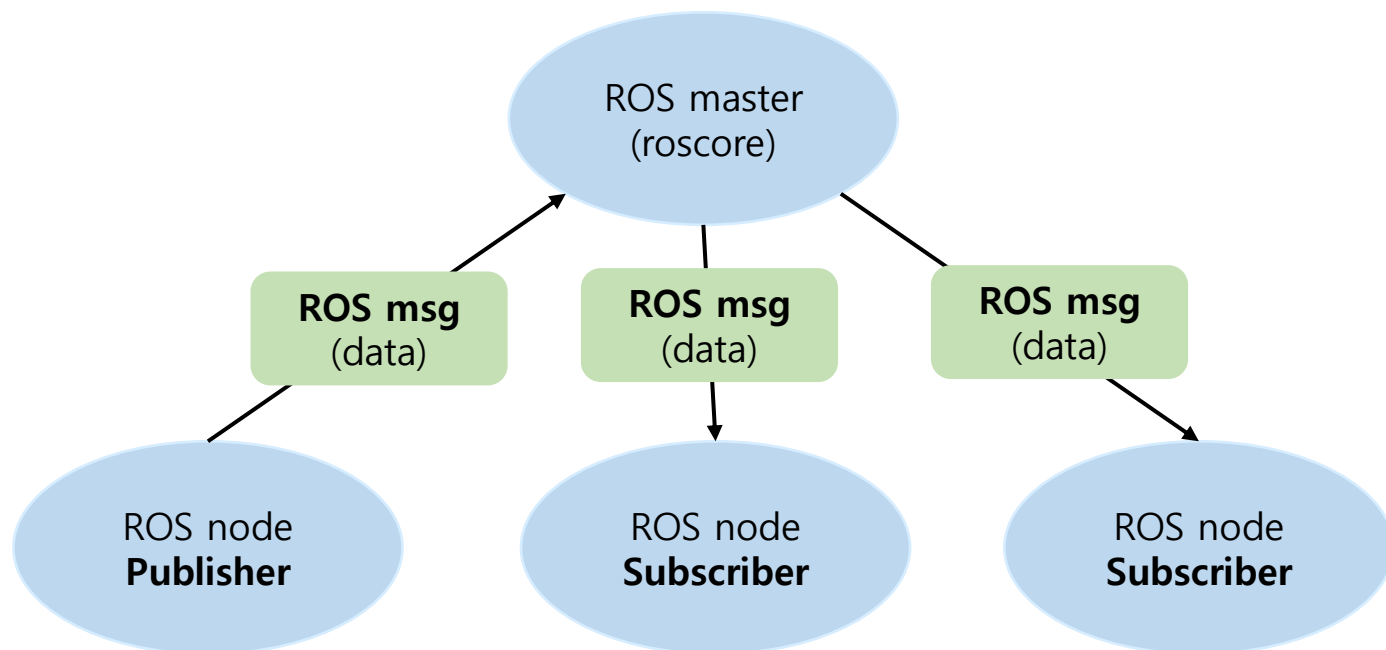
- In RDA(실시간 데이터 해석) course,
- We are preparing CUDA coding practice.
- We need **real time data** (streaming, in sequence)

Our data is rosbag streaming

ROS



NVIDIA Jetson Nano



01 Get into Environment

- So ROS and GPU needed, how to get to there?



Personal PC



NVIDIA Jetson Nano



Docker container with ROS

Connect to rda wifi (nano is connected to rda)

```
ssh "student_00"@192.168.0."groutnumber"  
password : inha_rcv
```

!! Plz type like this -> ex. ssh student_01@198.168.0.100

00 : student number assigned to you (02~50)

```
sudo docker run -it --runtime nvidia --network host -v /home/dataset/./data --name "student_00" dustynv/ros:melodic-ros-base-l4t-r32.7.1
```

!!Only once!! This is how to make docker container

```
sudo docker start "student_00" && sudo docker exec -it "student_00" /bin/bash
```

!!Always after once running command above!!

To get into existing docker container

!! Plz type like this ->

ex. sudo docker start student_01 &&

sudo docker exec -it student_01 /bin/bash

```
nvcc --version # command to check nvidia gpu  
rostopic list # check ros msg names
```

Works because we connected nano's gpu to docker container

01 Exit from Environment

- How to get out back to your computer's terminal? Exit



Docker container with ROS



NVIDIA Jetson Nano



Personal PC

```
exit
```

Leave docker environment

But generated docker container is still alive. Check it with `sudo docker ps -a` # shows list of docker containers

You can use it again with command `sudo docker start "student_00" && sudo docker exec -it "student_00" /bin/bash`

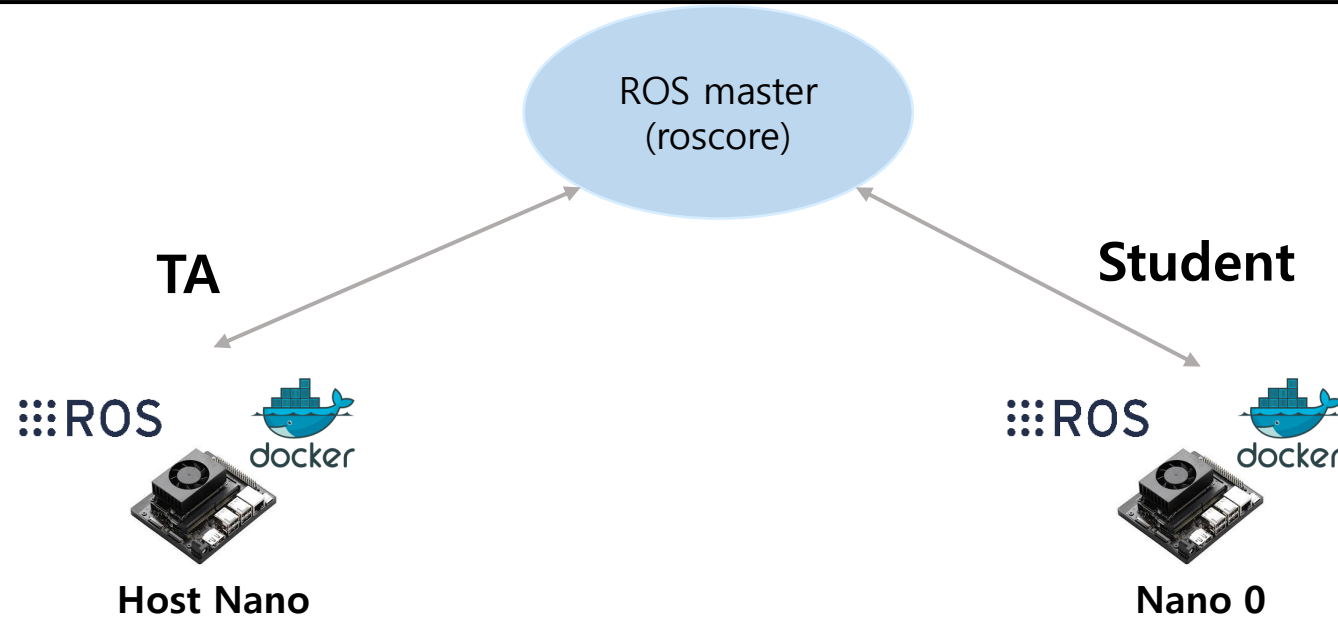
```
exit
```

Cut ssh connection

Get out of Nano

Back to my terminal

02 About Real-time ROS Data



Edit ~/.bashrc to share ROS master

Make and share ROS master

```
roscore # turn ROS master on (it's not your work)
```

```
nano ~/.bashrc
```

!! Plz type like this ->

ex. export ROS_MASTER_URI=http://192.168.0.106:11311

Add `export ROS_MASTER_URI=http://"master_ip":11311` in .bashrc

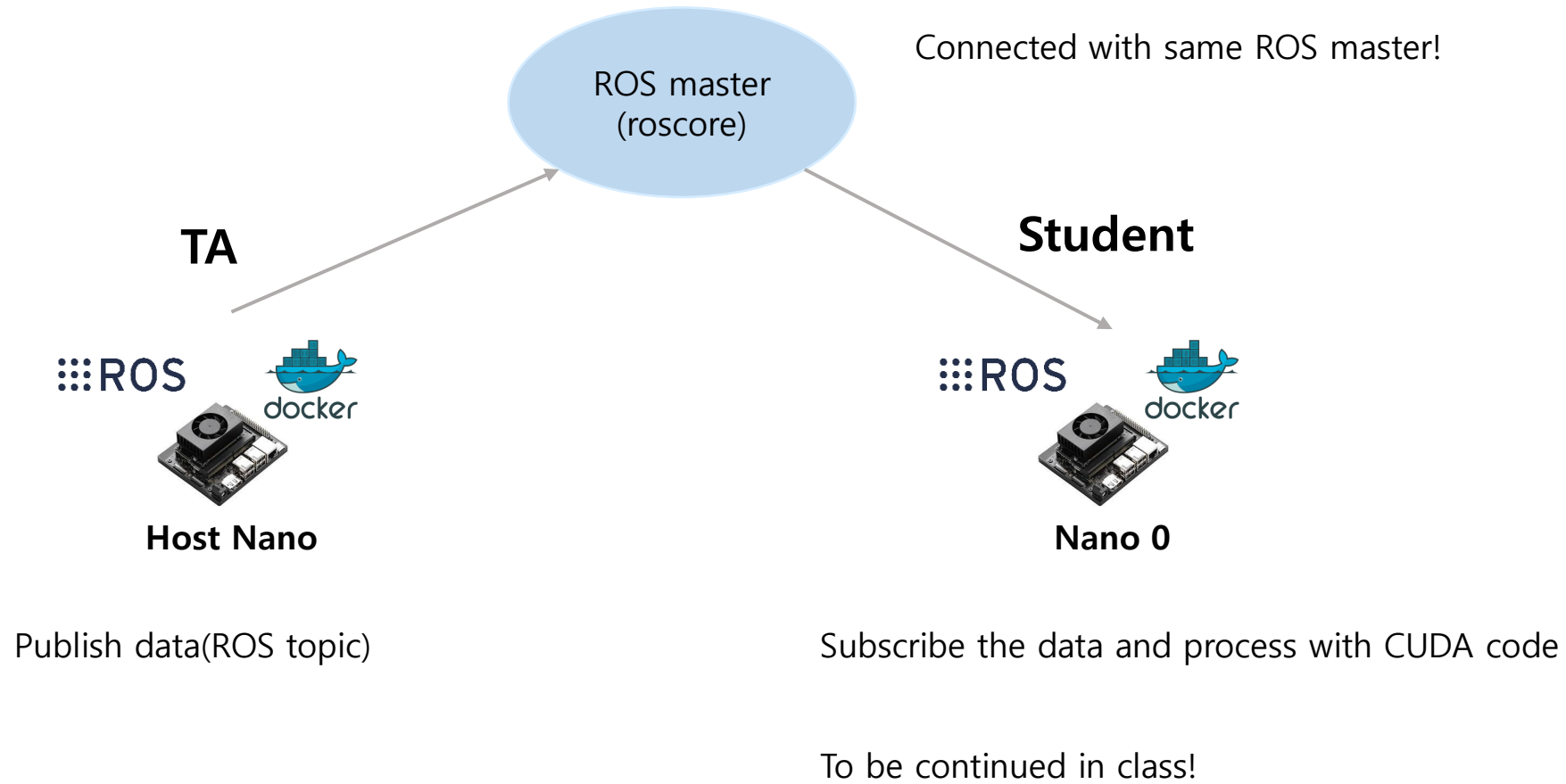
Host Nano IP

Add `source /opt/ros/melodic/setup.bash` in .bashrc

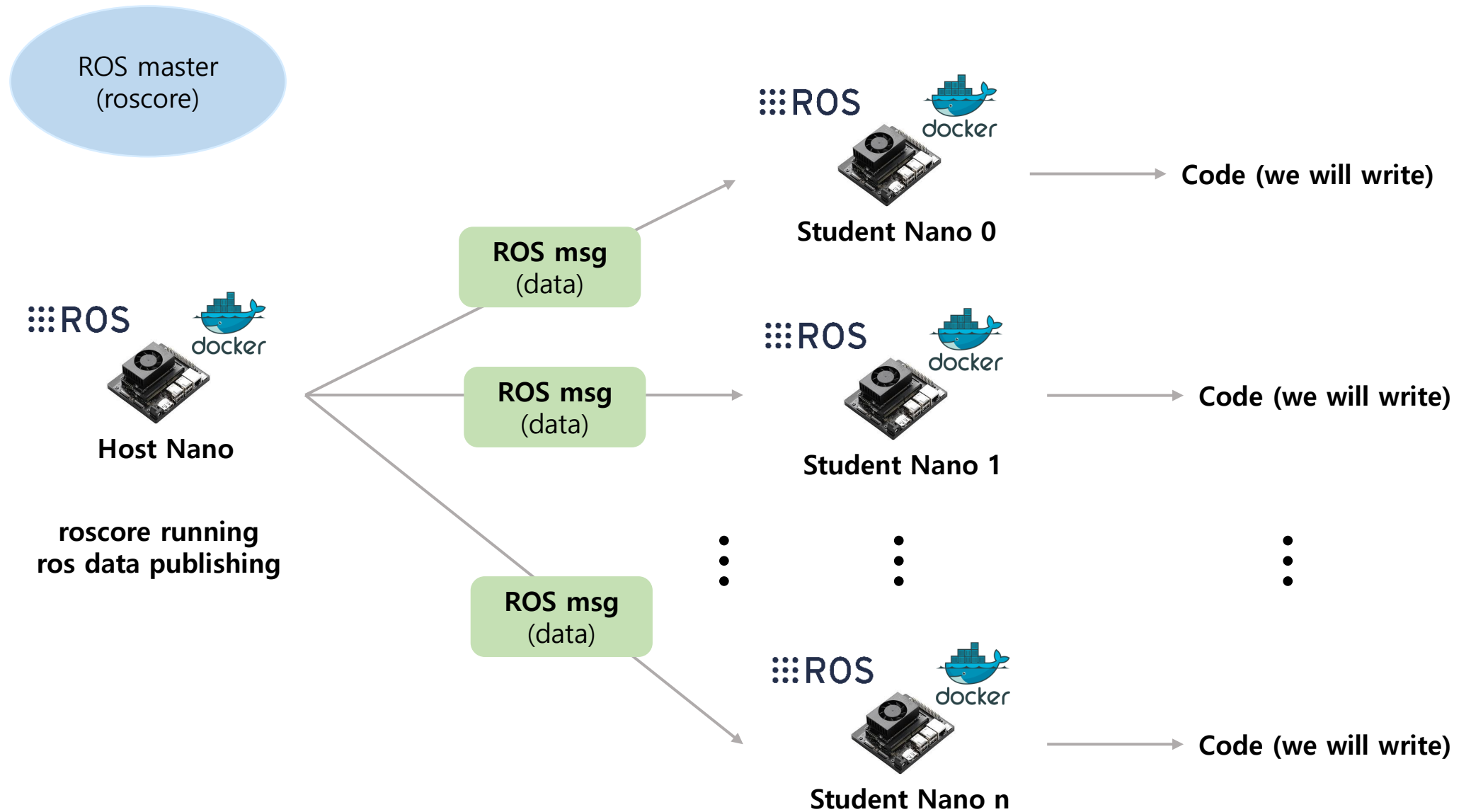
!! when saving Plz type "Ctrl+x+yes -> Ctrl+x"

```
source ~/.bashrc
```

02 About Real-time ROS Data



02 About Real-time ROS Data



03 Command Explanation (in prev pages)

```
ssh "student_00"@192.168.0."groutnumber"
```

The user name

Target Nano's IP

```
sudo docker run -it --runtime nvidia --network host -v /home/dataset/./data --name "student_00" dustynv/ros:melodic-ros-base-l4t-r32.7.1
```

Let docker container use GPU if you have

Container name

Docker image

This command makes docker container from docker image. It is concept of docker. If interested, search google it is simple and useful. Once container made we can re-use it.

```
sudo docker ps -a # shows list of docker containers
```

You can check container you made here.

If **student_00** found, run command below

```
sudo docker start "student_00" && sudo docker exec -it "student_00" /bin/bash
```

Start container cause you stopped with **exit**

Get into container's terminal

03 Command Explanation (in prev pages)

```
nvcc --version # command to check nvidia gpu  
rostopic list # check ros msg names
```

`nano ~/.bashrc` ~/.bashrc == absolute path to file
 ~ is special path
 .bashrc is a file and it is having environment settings in it
 . means that it is hidden file

Add `export ROS_MASTER_URI=http://"master_ip":11311` in .bashrc Connecting computers with one ros master to communicate (data or ...)

Add `source /opt/ros/melodic/setup.bash` in .bashrc Manually apply ros setting file (similar with source ~/.bashrc)

!! when saving Plz type "Ctrl+x+yes -> Ctrl+x"

`source ~/.bashrc` Manually apply new ~/.bashrc to the terminal